

Find the expression of work done in an isothermal reversible process for a gas obeying the equation of state as follows  $p(V-nb) = nRT$  where  $b$  is constant.

$$P_{op} = P_{in} - dp$$

$$dW_{rev} = P_{op} dv$$

$$= (P_{in} - dp) dv$$

$$= P_{in} dv - dp dv \quad dp \rightarrow 0$$

$$= P_{in} dv$$

$$P = \frac{nRT}{V-nb}$$

$$dW_{rev} = \frac{nRT}{V-nb} dv$$

$$W = \int_{V_1}^{V_2} \frac{nRT}{V-nb} dv$$

$$= nRT \int_{V_1}^{V_2} \frac{dv}{V-nb}$$

$$= nRT [\ln V - nb]_{V_1}^{V_2}$$

$$= nRT \ln \left( \frac{V_2 - nb}{V_1 - nb} \right)$$

$V_1 = \text{initial vol}$   
 $V_2 = \text{final vol.}$

Establish the following relation  $C_p - C_v = \left[ P + \left( \frac{\partial U}{\partial V} \right)_T \right] \left[ \frac{\partial V}{\partial T} \right]_P$   
and from there establish the following relation  $b/n$   
 $C_p$  and  $C_v$  for ideal gas

$$C_p = \left( \frac{\partial H}{\partial T} \right)_P \quad C_v = \left( \frac{\partial U}{\partial T} \right)_V$$

$$= \left[ \frac{\partial (U + PV)}{\partial T} \right]_P$$

$$= \left( \frac{\partial U}{\partial T} \right)_P + P \left( \frac{\partial V}{\partial T} \right)_P$$

$$C_p = \left( \frac{\partial U}{\partial T} \right)_V + \left( \frac{\partial U}{\partial V} \right)_T \left( \frac{\partial V}{\partial T} \right)_P + P \left( \frac{\partial V}{\partial T} \right)_P$$

$$\Rightarrow C_p = C_v + \left( \frac{\partial U}{\partial V} \right)_T \left( \frac{\partial V}{\partial T} \right)_P + P \left( \frac{\partial V}{\partial T} \right)_P$$

$$U = f(T, V)$$

$$\partial U = \left( \frac{\partial U}{\partial T} \right)_V \partial T + \left( \frac{\partial U}{\partial V} \right)_T \partial V$$

$$\left( \frac{\partial U}{\partial T} \right)_P = \left( \frac{\partial U}{\partial T} \right)_V + \left( \frac{\partial U}{\partial V} \right)_T \left( \frac{\partial V}{\partial T} \right)_P$$

$$C_p - C_v = \left[ P + \left( \frac{\partial U}{\partial V} \right)_T \right] \left( \frac{\partial V}{\partial T} \right)_P$$

For ideal gas  $U = U(T)$

$$\left( \frac{\partial U}{\partial V} \right)_T = 0$$

$$C_p - C_v = P \left( \frac{\partial V}{\partial T} \right)_P$$

$$PV = nRT$$

$$= P \frac{nR}{P} \quad \left( \frac{dV}{dT} \right)_P = \frac{nR}{P}$$

$$\frac{T \Delta n}{dT - V} = nR$$

For one mol

$$C_p - C_v = R$$

One mole of a monoatomic ideal gas ( $C_v = \frac{3}{2}R$ ) is expanded adiabatically and reversibly from 1 atm, 273K to final pressure 0.5 atm.

Find final temperature, work done and enthalpy change

$$C_p - C_v = R$$

$$C_p = \frac{3}{2}R + R = \frac{5}{2}R$$

$$\gamma = \frac{C_p}{C_v} = \frac{5}{3}$$

$$PV^\gamma = C$$

$$P_1^{1-\gamma} T_1^\gamma = P_2^{1-\gamma} T_2^\gamma$$

$$\Rightarrow T_2 = T_1 \left( \frac{P_1}{P_2} \right)^{\frac{1-\gamma}{\gamma}}$$

$$= 273 \left( \frac{1}{1/2} \right)^{-\frac{2/3}{5/3}}$$

$$= 273 (2)^{-\frac{2}{5}}$$

$$= 273 \left( \frac{1}{2} \right)^{\frac{2}{5}} = 206.89 \text{ K}$$

$$\Delta T = -66.11$$

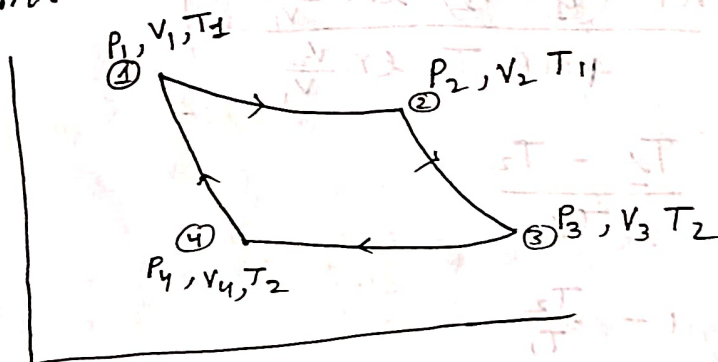
work done

$$W = \frac{nR\Delta T}{1-\gamma}$$
$$= \frac{1 \times 8.314 \times -66.11}{-2/3}$$
$$= 824.4 \text{ J}$$

Enthalpy Change  $\Delta H$

$$\Delta H = nC_p \Delta T$$
$$= 1 \times \frac{5}{2} R \times 66.11$$
$$= -1371.78 \text{ J}$$

Calculated the amount of work involved in each step of Carnot cycle and from there find out the expression of efficiency of Carnot engine with clearly depicted P-V diagram.



1 → 2 Isothermal expansion

$$\Delta U = 0$$

$$W_{12} = -nRT_1 \ln \frac{V_2}{V_1}$$

$$q_1 = -W = nRT_1 \ln \frac{V_2}{V_1}$$

2 → 3 adiabatic expansion

$$W_{23} = \Delta U$$

$$= nC_v \Delta T$$

$$= nC_v (T_2 - T_1)$$

$$q = 0$$

→ ⑤

→ (C)

$$T \Delta Q_{\text{rev}} = H_0$$

$$\textcircled{2} W_{\text{Total}} = W_{12} + W_{23} + W_{34} + W_{41}$$

$$T_2 \ln \frac{v_2}{v_3} + nC_v(T_1 - T_2)$$

*[Faint handwritten notes at the bottom of the page, possibly "P-V diagram"]*

$$\frac{T_1 - T_2}{T_1} + nR T_1 \ln \frac{v_2}{v_1}$$

$\frac{1}{2} \times 10 \times 10 = 50$

•  $f$  function

$$\Delta S = \oint \frac{dq_{rev}}{T} = \frac{q_1}{T_1} + \frac{q_2}{T_2}$$

$$V_1 V_3 = V_2 V_4$$

$$\frac{V_3}{V_4} = \frac{V_2}{V_1}$$

$$= 0$$

In a close path  $\Delta S$  is zero

that's mean entropy is a state function.

Calculate entropy change for 3 moles of a diatomic ideal gas which is heated and compressed from 298K and 1 bar to 398 at 2 bar ( $C_p = \frac{7}{2} R$ )

$$\Delta S = n C_v \ln \left( \frac{T_2}{T_1} \right) + n R \ln \left( \frac{V_2}{V_1} \right)$$

$$\frac{V_2}{V_1} = \frac{T_2 P_1}{T_1 P_2}$$

$$= n C_v \ln \frac{T_2}{T_1} + n R \ln \left( \frac{T_2 P_1}{T_1 P_2} \right)$$

$$= n (C_v + R) \ln \frac{T_2}{T_1} + n R \ln \left( \frac{P_1}{P_2} \right)$$

$$= n C_p \ln \frac{T_2}{T_1} + n R \ln \left( \frac{P_1}{P_2} \right)$$

$$P_1 = 1 \quad P_2 = 2 \quad n = 3 \quad C_p = \frac{7}{2} R$$

$$T_1 = 298 \quad T_2 = 398$$

$$\Delta S = 3 \times \frac{7}{2} R \ln \frac{398}{298} + 3 \times R \ln \left( \frac{1}{2} \right)$$

$$= 7.97 \text{ J}$$

"Adiabatic process is an isentropic process"  
- justify or criticize

$$ds = \frac{dQ_{rev}}{T}$$

Reversible adiabatic process

For adiabatic process:  $dQ = 0$

For reversible process  $dQ_{rev} = 0$

$$ds = \frac{0}{T} = 0$$

entropy remain constant

Reversible adiabatic process is isentropic

For irreversible process:

$$ds > \frac{dQ}{T}$$

$$ds > 0$$

Entropy increases:

So irreversible adiabatic process is NOT isentropic

Assuming ideal behavior, calculate the total change in entropy for 5 gm-mole of water vapour initially at a temperature of  $227^\circ\text{C}$  cooled down to liq. water at  $27^\circ\text{C}$  at 1 atm pressure (heat capacity of water =  $1 \text{ cal/gm}$ , heat capacity of water vapour =  $0.4 \text{ cal/gm}$ , latent heat of vaporization of water =  $540 \text{ cal/gm}$ )

No. of mol = 5 gm-mol

Molar mass =  $18 \text{ g/mol}$

$$m = 5 \times 18 = 90 \text{ g}$$

$$T_1 = 227^\circ\text{C} = 500\text{K}$$

$$T_2 = 100^\circ\text{C} = 373\text{K}$$

$$T_3 = 27^\circ\text{C} = 300\text{K}$$

$$C_{\text{steam}} = 0.4 \text{ cal/gK} \quad L = 540 \text{ cal/g}$$

$$C_{\text{water}} = 1 \text{ cal/gK}$$

$$\Delta S_1 = m C_{\text{steam}} \ln\left(\frac{T_2}{T_1}\right)$$

$$= 100 \times 0.4 \ln\left(\frac{373}{500}\right)$$

$$= -10.54 \text{ cal/K}$$

$$\Delta S_2 = -\frac{mL}{T_2}$$

$$= -\frac{100 \times 540}{373}$$

$$= -130.29 \text{ cal/K}$$

$$\Delta S_3 = m C_{\text{water}} \ln\left(\frac{T_3}{T_2}\right)$$

$$= 100 \times 1 \times \ln\left(\frac{300}{373}\right)$$

$$= -19.60 \text{ cal/K}$$

$$\Delta S = \Delta S_1 + \Delta S_2 + \Delta S_3$$

$$= -160.43 \text{ cal/K}$$

$$= -673.81 \text{ J/K}$$

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A car engine with a power output of 50 kW has a thermal efficiency of 25 percent. Determine the fuel consumption rate (kg/sec) of this car if the fuel has a heating value of 40,000 kJ/kg

$$\eta = 0.25 \quad P_{out} = 50 \text{ kW} = 50 \text{ kJ/s}$$

$$\eta = \frac{\text{Power output}}{\text{Heat input per sec}}$$

$$0.25 = \frac{50}{H}$$

$$H = 200 \text{ kJ/s}$$

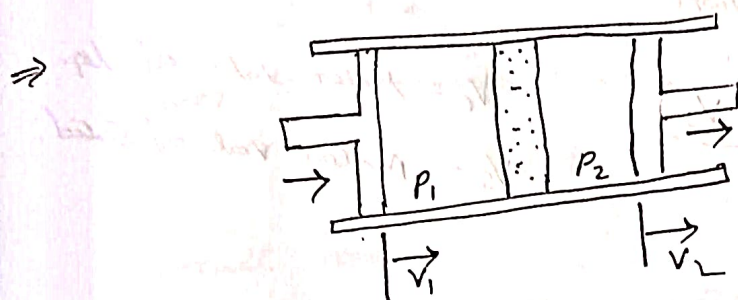
$$\text{Heating value} = 40,000 \text{ kJ/kg}$$

$$\text{Fuel consumption rate} = \frac{\text{Heat input per sec}}{\text{Heating value}}$$

$$= \frac{200}{40,000} \text{ kg/s}$$

$$= 0.005 \text{ kg/s}$$

Draw a schematic diagram of Joule-Thomson experiment.  
 why Joule-Thomson experiment is called "iso-enthalpic process"? what is inversion temperature.



adiabatic  $Q=0$   
 no external work  $w=0$

$$P_1 > P_2$$

$V_1$  decrease in volume

$V_2$  increase in volume

$$\text{Work on the system} = P_1 V_1$$

$$\text{Work by the system} = P_2 V_2$$

$$\text{net work done on the system} = P_1 V_1 - P_2 V_2 = \Delta U$$

$$\Rightarrow P_1 V_1 - P_2 V_2 = U_2 - U_1$$

$$\Rightarrow U_1 + P_1 V_1 = U_2 + P_2 V_2$$

$$\Rightarrow H_1 = H_2$$

$$\Rightarrow \Delta H = 0$$

So it is isenthalpic process

using the Clausius-Clayperon equation, find at what pressure water will boil at  $98^\circ\text{C}$ ?

$$\ln \frac{P_2}{P_1} = -\frac{L}{R} \left( \frac{1}{T_2} - \frac{1}{T_1} \right)$$

$$T_1 = 373 \text{ K}$$

$$T_2 = 98^\circ\text{C} = 371 \text{ K}$$

$$\Rightarrow \ln \left( \frac{P_2}{1} \right) = -\frac{40.8 \times 10^3}{8.314} \left( \frac{1}{371} - \frac{1}{373} \right)$$

$$L = 540 \text{ cal/g}$$

$$= 540 \times 4.2 \times 18 \text{ J/mol}$$

$$= 40.8 \times 10^3 \text{ J/mol}$$

$$R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$$

$$P_1 = 1 \text{ atm}$$

$$P_2 = ?$$

$$\Rightarrow P_2 \approx 0.93 \text{ atm.}$$

Effect of change of pressure on melting point of ice and solid  $\text{CO}_2$  shows opposite trend - explain.

$\Rightarrow$  by Clapeyron equation

$$\frac{dT}{dP} = \frac{T(V_L - V_S)}{L}$$

$V_L$  = Molar Vol. of liq.  
 $V_S$  = Molar vol of Solid

For Ice ( $\text{H}_2\text{O}$ )

Ice has an open structure  $\rightarrow$  Volume decreases on melting.

$$\text{so } V_L < V_S \Rightarrow V_L - V_S < 0$$

$$\frac{dT}{dP} < 0$$

Melting point decreases with increase in pressure.

For solid  $\text{CO}_2$  (Dry Ice)

Solid  $\text{CO}_2$  is more compact  $\rightarrow$  Volume increases on melting

$$\text{so, } V_L > V_S \Rightarrow V_L - V_S > 0$$

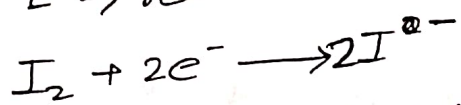
$$\frac{dT}{dP} > 0$$

Melting point increases with increase in pressure.

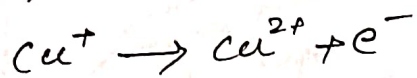
a) Given  $E^\circ(\text{Cu}^{2+}/\text{Cu}^+) = +0.15\text{V}$  and  $E^\circ(\text{I}_2/\text{I}^-) = 0.54\text{V}$   
Write down the possible reaction from these  $E^\circ$  values. How can you explain the formation of iodine vapour from those standard redox potential values.

For the reaction  $E_{\text{cell}}^{\circ} = +ve$  for spontaneous rxn

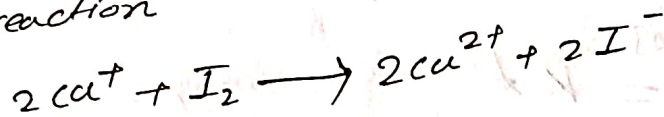
Higher  $E^{\circ} \rightarrow$  reduction



Lower  $E^{\circ} \rightarrow$  reverse (oxidation)



Overall reaction



$$E_{\text{cell}}^{\circ} = 0.54 - 0.15 = 0.39 \text{ V}$$

so it is spontaneous

For a two step consecutive reaction, what is meant by steady-state approximation, when is it valid?

What is the utility of it?

$\Rightarrow$  In a two step consecutive reaction



Intermediate  $I$  is highly reactive

Its concentration remains very small and nearly constant during reaction.

$$\frac{d[I]}{dt} \approx 0$$

Validity

When intermediate is formed and consumed rapidly

when its concentration is much smaller than reactants.

Utility  
Simplifies rate eq. by eliminating intermediate concn.

Helps to derive an overall rate law in terms of reactant only

makes solving complex mechanism mathematically easier.

Show that half-life of a second order reaction is dependent on initial concentration of reactant.

$$\Rightarrow -\frac{d[A]}{dt} = k[A]^2$$

$$-\int_{[A]_0}^{[A]} \frac{d[A]}{[A]^2} = k \int_0^t dt$$

$$\Rightarrow \left[ \frac{1}{[A]} \right]_{[A]_0}^{[A]} = kt$$

$$\Rightarrow \frac{1}{[A]} - \frac{1}{[A]_0} = kt$$

$$\text{at } t_{\frac{1}{2}} \quad [A] = \frac{[A]_0}{2}$$

$$\Rightarrow \frac{2}{[A]_0} - \frac{1}{[A]_0} = kt_{\frac{1}{2}}$$

$$\Rightarrow \frac{1}{[A]_0} = kt_{\frac{1}{2}} \Rightarrow t_{\frac{1}{2}} = \frac{1}{k[A]_0}$$

So  $t_{\frac{1}{2}}$  dependent on initial concentration

For a reversible reaction  $A \rightleftharpoons B$  find out the equilibrium concentration of A and B when the ratio of forward to backward is 2 with initial concentration of A as  $A_0$ .



$$[A] = A_0 - x \quad [B] = x$$

$$K_{eq} = \frac{[B]}{[A]} = \frac{K_f}{K_b}$$

$$\Rightarrow \frac{x}{A_0 - x} = 2$$

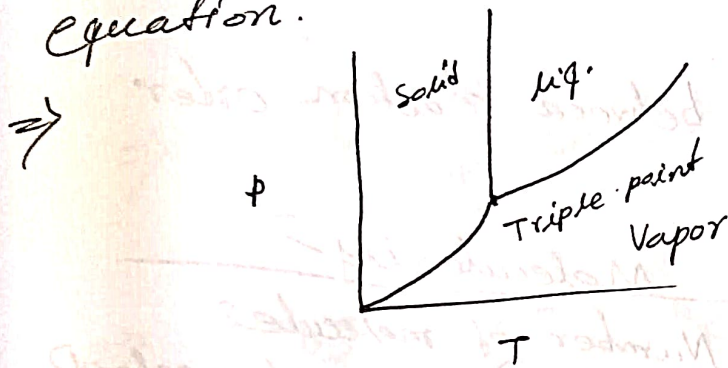
$$\Rightarrow 2x = 2A_0 - 2x$$

$$\Rightarrow 3x = 2A_0$$

$$\Rightarrow x = \frac{2}{3} A_0$$

$$[B] = \frac{2}{3} A_0 \quad [A] = A_0 - \frac{2}{3} A_0 = \frac{1}{3} A_0$$

Derive the expression of the slope of  $p$  vs  $T$  phase equilibrium - diagram using Clausius - Clapeyron equation.



At phase equilibrium: chemical potentials are equal

$$dG = V dp - S dT$$

For two phase

$$V_2 dp - S_2 dT = V_1 dp - S_1 dT$$

$$\Rightarrow (V_2 - V_1) dp = (S_2 - S_1) dT$$

$$\Rightarrow \frac{dp}{dT} = \frac{S_2 - S_1}{V_2 - V_1}$$

$$\Rightarrow \frac{dp}{dT} = \frac{T(S_2 - S_1)}{T(V_2 - V_1)}$$

$$L = T(S_2 - S_1) = \text{Latent heat}$$

$$\Rightarrow \frac{dp}{dT} = \frac{L}{T(V_2 - V_1)}$$

Calculate the entropy change for 3 mole of an ideal diatomic gas which is heated and compressed from  $(25^\circ\text{C}, 1\text{ bar})$  to  $(210^\circ\text{C}, 3\text{ bar})$   $\bar{C}_V = \frac{5}{2}R$

$$\Delta S = n C_V \ln \frac{T_2}{T_1} + n R \ln \frac{V_2}{V_1}$$

$$pV = nRT$$

$$\frac{V_2}{V_1} = \frac{T_2 P_1}{T_1 P_2}$$

$$n = 3 \text{ mol} \quad T_1 = 25^\circ\text{C} = 298\text{K}$$

$$T_2 = 210^\circ\text{C} = 483\text{K}$$

$$P_1 = 1\text{ bar}$$

$$P_2 = 3\text{ bar}$$

$$C_V = \frac{5}{2}R$$

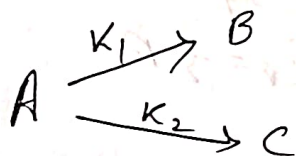
$$\Delta S = 3 \times \frac{5}{2} R \ln \frac{483}{298} + 3R \ln \left( \frac{483.1}{298.5} \right)$$

$$= 14.75 \text{ J/K}$$

What are the differences between reaction order and molecularity?

| Order of Reaction                                | Molecularity                                        |
|--------------------------------------------------|-----------------------------------------------------|
| Sum of powers of concentration terms in rate law | Number of molecules colliding in an elementary step |
| Determine experimentally                         | Determined from mechanism (theory)                  |
| Can be zero, fraction or integer                 | always a whole number (1, 2, ...)                   |
| Applies to overall reaction                      | Applies only to elementary reaction                 |

For a parallel reaction of following kind, reduce the expression of concentration profile with time for  $[A(t)]$ ,  $[B(t)]$ ,  $[C(t)]$  and show that the ratio  $\frac{[C(t)]}{[B(t)]}$  for  $t \rightarrow \infty$  is always constant. Consider both reactions are of first order.



$$\Rightarrow -\frac{d[A]}{dt} = (k_1 + k_2)[A]$$

$$\Rightarrow \ln \frac{[A]_0}{[A]} = (k_1 + k_2)t$$

$$\Rightarrow \frac{[A]}{[A]_0} = e^{-(k_1 + k_2)t}$$

$$\Rightarrow [A] = [A]_0 e^{-(k_1 + k_2)t}$$

Concentration of  $B(t)$

$$\frac{d[B]}{dt} = k_1[A]$$

$$\Rightarrow \frac{d[B]}{dt} = k_1[A] \cdot e^{-(k_1 + k_2)t}$$

$$\Rightarrow d[B] = k_1[A]_0 e^{-(k_1+k_2)t} dt$$

$$\Rightarrow [B] = \frac{k_1 [A]_0}{-(k_1 + k_2)} e^{-(k_1 + k_2)t} + C \rightarrow \text{---}$$

at  $x=0$   $[B]=0$

$$\Rightarrow 0 = - \frac{k_1 [A]_0}{(k_1 + k_2)} + C$$

so eq. (1)

so eq. (1)

$$[B] = \frac{k_1 [A]_0}{k_1 + k_2} (1 - e^{-(k_1 + k_2)t})$$

Similarly

$$\frac{d[A]}{dt} = k_2 [A]$$

$$\text{So } [C] = \frac{k_2 [A]_0}{k_1 + k_2} (1 - e^{-(k_1 + k_2)t})$$

So ratio

$$\frac{[C]}{[B]} = \frac{k_2}{k_1} \approx \text{constant for } t \gg 0$$

Calculate the entropy change when 10 gm of Neon is heated from  $27^\circ\text{C}$  to  $227^\circ\text{C}$  at constant Volume ( $C_v = 3 \text{ cal/mole}$ )

$\Rightarrow$  Molar mass of Ne = 20 g/mol

$$n = \frac{10}{20} = 0.5 \quad T_1 = 27^\circ\text{C} = 300 \text{ K}$$

$$T_2 = 227^\circ\text{C} = 500 \text{ K}$$

$$C_v = 3 \text{ cal/mol}\cdot\text{K}$$

$$\Delta S = n C_v \ln\left(\frac{T_2}{T_1}\right) + n R \ln\frac{V_2}{V_1}$$

$V = \text{Constant}$

$$\Delta S = n C_v \ln\left(\frac{T_2}{T_1}\right)$$

$$= \frac{1}{2} \times 3 \ln\left(\frac{500}{300}\right)$$

$$= 0.766 \text{ cal/K}$$

Explain why two ice cube get fused when pressed together

$$\Rightarrow \text{For ice } V_L < V_S \Rightarrow \frac{dT}{dP} < 0$$

increasing pressure lowers melting point

When ice cubes are pressed  $\rightarrow$  pressure increases at contact

Melting point decreases  $\rightarrow$  ice melts locally

When pressure is removed  $\rightarrow$  water refreezes

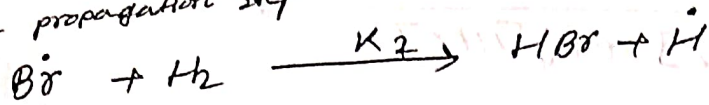
so ice cube get fused.

For the reaction  $\text{H}_2 + \text{Br}_2 \rightarrow 2\text{HBr}$ , shows that the order is  $\frac{1}{2}$  w.r.t.  $\text{Br}_2$  at the initial stage of the reaction.

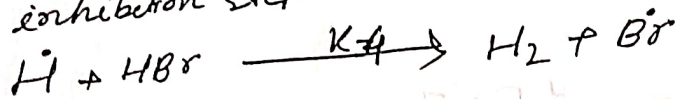
$\Rightarrow$  Chain initiation step



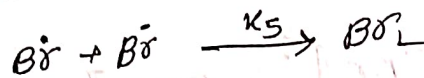
Chain propagation step



Chain inhibition step



Chain Termination step



$$\frac{d[\text{HBr}]}{dt} = k_2[\text{Br}\cdot][\text{H}_2] + k_3[\text{H}\cdot][\text{Br}_2] - k_4[\text{H}\cdot][\text{HBr}] \rightarrow (1)$$

$$\frac{d[\text{H}\cdot]}{dt} = 0 = k_2[\text{Br}\cdot][\text{H}_2] - k_3[\text{H}\cdot][\text{Br}_2] - k_4[\text{H}\cdot][\text{HBr}]$$

$$\Rightarrow [\text{H}\cdot] = \frac{k_2[\text{Br}\cdot][\text{H}_2]}{k_3[\text{Br}_2] + k_4[\text{HBr}]} \rightarrow (2)$$

$$\frac{d[\text{Br}\cdot]}{dt} = 0 = 2k_1[\text{Br}_2] - k_2[\text{Br}\cdot][\text{H}_2] + k_3[\text{H}\cdot][\text{Br}_2] + k_4[\text{H}\cdot][\text{HBr}] - 2k_5[\text{Br}\cdot]^2$$

$$0 = 2k_1[\text{Br}_2] - 2k_5[\text{Br}\cdot]^2 - (k_2[\text{Br}\cdot][\text{H}_2] - k_3[\text{H}\cdot][\text{Br}_2] - k_4[\text{H}\cdot][\text{HBr}])$$

$$\Rightarrow 0 = 2k_1[\text{Br}_2] - 2k_5[\text{Br}\cdot]^2$$

$$\Rightarrow 2k_1[\text{Br}_2] = 2k_5[\text{Br}\cdot]^2$$

$$\Rightarrow [\text{Br}\cdot] = \left(\frac{k_1}{k_5}\right)^{\frac{1}{2}} [\text{Br}_2]^{\frac{1}{2}} \rightarrow (3)$$

$$[\dot{H}] = \frac{k_2 [Br] [H_2]}{k_3 [Br_2] + k_4 [HBr]}$$

$$[\dot{H}] = \frac{k_2 \left(\frac{k_1}{k_5}\right)^{\frac{1}{2}} [Br_2]^{\frac{1}{2}}}{k_3 [Br_2] + k_4 [HBr]} \quad \text{put - (3)}$$

$$\frac{d[HBr]}{dt} = k_2 [Br] [H_2] + k_3 [\dot{H}] [Br_2] - k_4 [\dot{H}] [HBr]$$

$$= k_2 \left(\frac{k_1}{k_5}\right)^{\frac{1}{2}} [Br_2]^{\frac{1}{2}} [H_2]$$

$$+ k_3 \frac{k_2 \left(\frac{k_1}{k_5}\right)^{\frac{1}{2}} [Br_2]^{\frac{1}{2}}}{k_3 [Br_2] + k_4 [HBr]} \cdot [Br_2]$$

$$- k_4 \frac{k_2 \left(\frac{k_1}{k_5}\right)^{\frac{1}{2}} [Br_2]^{\frac{1}{2}}}{k_3 [Br_2] + k_4 [HBr]} \cdot [HBr]$$

$$= k_2 \left(\frac{k_1}{k_5}\right)^{\frac{1}{2}} [Br_2]^{\frac{1}{2}} [H_2]$$

$$\left( 1 + \frac{k_3 [Br_2]}{k_3 [Br_2] + k_4 [HBr]} - \frac{k_4 [HBr]}{k_3 [Br_2] + k_4 [HBr]} \right)$$

$$= k_2 \left(\frac{k_1}{k_5}\right)^{\frac{1}{2}} [Br_2]^{\frac{1}{2}} [H_2]$$

$$\left( \frac{k_3 [Br_2] + k_4 [HBr] + k_3 [Br_2] - k_4 [HBr]}{k_3 [Br_2] + k_4 [HBr]} \right)$$

$$= k_2 \left(\frac{k_1}{k_5}\right)^{\frac{1}{2}} [Br_2]^{\frac{1}{2}} [H_2] \left( \frac{2}{1 + \frac{k_4 [HBr]}{k_3 [Br_2]}} \right)$$

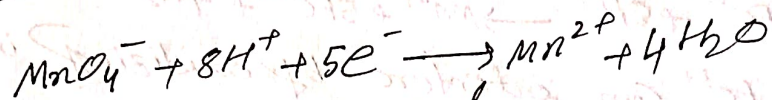
$$= K' \frac{[Br_2]^{\frac{1}{2}} [H_2]}{1 + K'' \frac{[HBr]}{[Br_2]}}$$

at initial stage  $[HBr] \approx 0$   
order. w.r.t.  $Br_2$  is  $\frac{1}{2}$

The oxidizing power of  $\text{MnO}_4^-$  varies with change of pH explain taking one example.

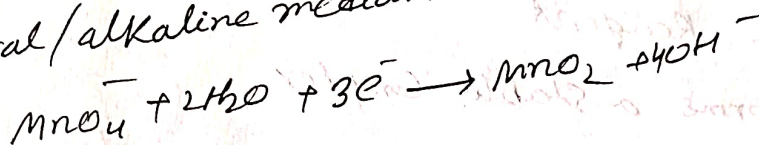
⇒ The reduction products of permanganate vary with pH

Acidic medium:



Strong oxidizing agent

Neutral/alkaline medium:



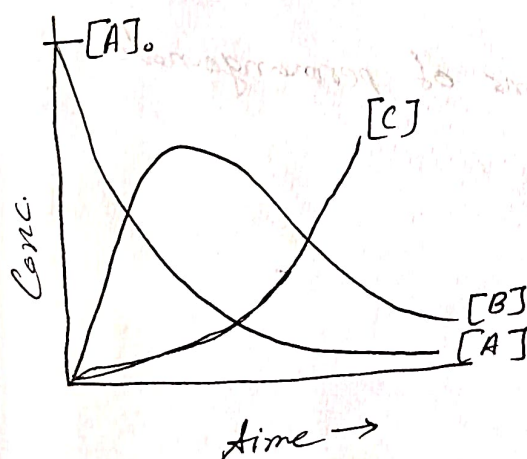
Weak oxidizing agent

as medium change pH change then accordingly oxidizing power change.

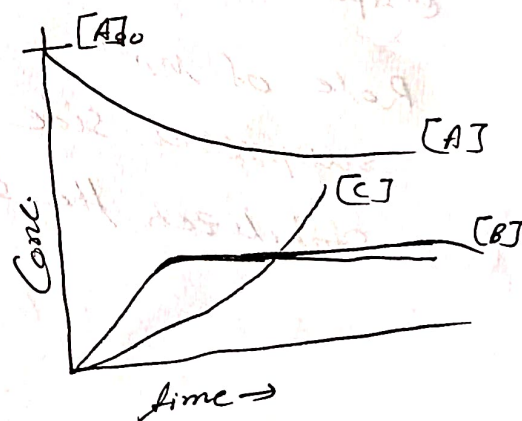
Draw only the graph for consecutive reaction:

$\text{A} \xrightarrow{k_1} \text{B} \xrightarrow{k_2} \text{C}$  with different relative magnitudes of two rate constant such as i)  $k_1 \gg k_2$  ii)  $k_1 \ll k_2$

$k_1 \gg k_2$

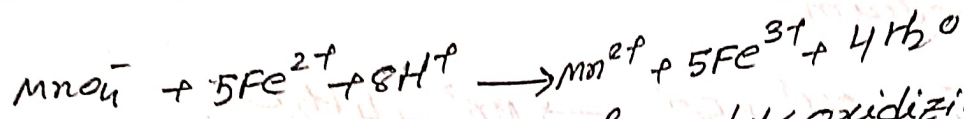


$k_1 \ll k_2$



During estimation of Fe(III) using permanganometry discuss the role of Zimmermann-Reinhardt reagent taking the help of Nernst equation.

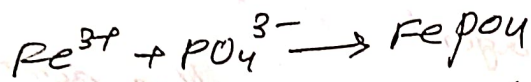
⇒ ZR reagent is a mixture of  $\text{MnSO}_4 + \text{H}_3\text{PO}_4 + \text{H}_2\text{SO}_4$



The formed  $\text{Fe}^{3+}$  can interfere by oxidizing indicator or causing side reaction.

Role of ZR Reagent

$\text{Fe}^{3+}$  forms a stable complex



Reduces free  $\text{Fe}^{3+}$  concentration

$$E = E^\circ_{\text{cell}} + \frac{0.059}{n} \log \frac{[\text{Fe}^{3+}]}{[\text{Fe}^{2+}]}$$

due to complexation  $[\text{Fe}^{3+}] \downarrow$

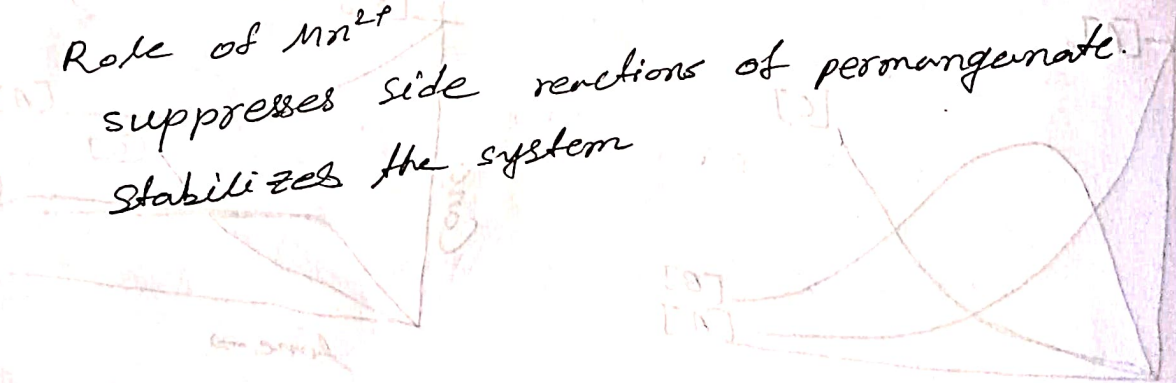
Hence, electrode potential decreases

This prevents  $\text{Fe}^{3+}$  from interfering ensures sharp endpoint.

Role of  $\text{Mn}^{2+}$

suppresses side reactions of permanganate.

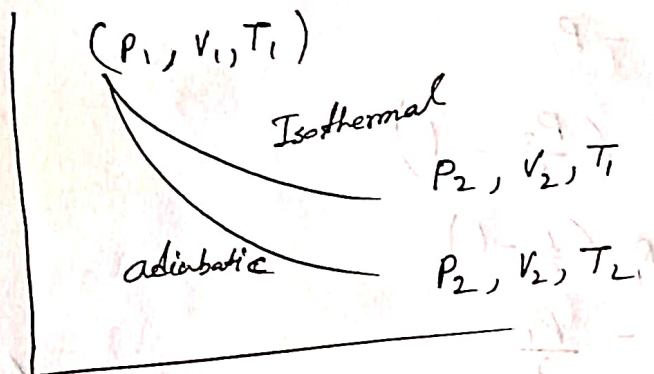
Stabilizes the system



Sketch the variation of pressure vs Volume curve for expansion of one mole of an ideal gas from initial  $(P_1, V_1, T_1)$  to final  $(P_2, V_2, T_2)$  for

I. Isothermal Condition  $(T_1 = T_2)$

II. Adiabatic Condition  $(T_1 > T_2)$



One mole of an ideal gas undergoes a reversible Carnot cycle with  $V_1 = 20 \text{ dm}^3$ ,  $V_2 = 40 \text{ dm}^3$  and  $T_1 = 300 \text{ K}$ . Let  $T_2 = 200 \text{ K}$  and  $C_{V,m} = \frac{3}{2} R$ . Calculate  $\Delta P$ ,  $\Delta H$ ,  $\Delta T$  and  $\Delta S$  for the isothermal expansion process given  $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$ .

⇒ Isothermal expansion of Gas

$$\Delta T = 0$$

$$H = H(T) \Rightarrow \Delta H = 0$$

$$\Delta S = nR \ln\left(\frac{V_2}{V_1}\right) + nC_V \ln\left(\frac{T_2}{T_1}\right) \quad T \text{ Constant}$$

$$= nR \ln \frac{V_2}{V_1}$$

$$= 1 \times 8.314 \times \ln\left(\frac{40}{20}\right)$$

$$= 5.76 \text{ J/K}$$

4.

$$P_1 = \frac{nRT}{V_1}$$

$$P_2 = \frac{nRT}{V_2}$$

$$\frac{P_2}{P_1} = \frac{V_1}{V_2} = \frac{1}{2}$$

$$\Delta P = P_2 - P_1$$

$$= P_1 \left( \frac{P_2}{P_1} - 1 \right)$$

$$= P_1 \left( \frac{1}{2} - 1 \right)$$

$$= -\frac{P_1}{2}$$

The Conversion of  $A \rightarrow B$  follow second order kinetics. If the concentration of A is increased to three time, how will it affect the overall rate of the reaction?

$$\text{Rate} = k[A]^2$$

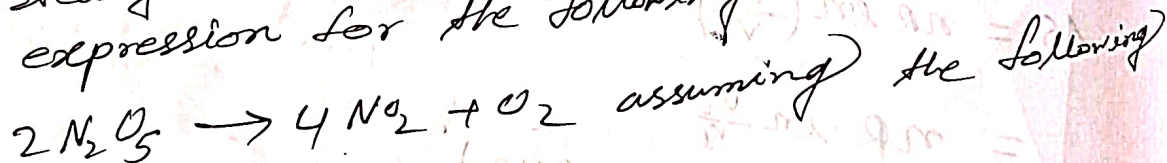
A is increase 3 times

$$\text{Rate}' = k[A]^2$$

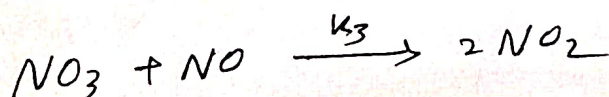
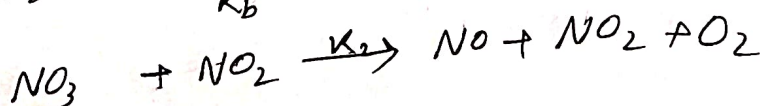
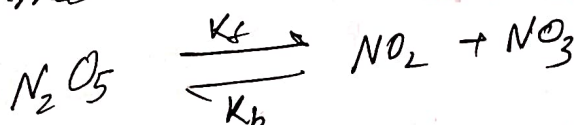
$$= 9 k[A]^2$$

$$= 9 \text{ Rate}$$

use steady state approximation to derive the rate expression for the following reaction



step mechanism



∴ show that the rate of formation of  $O_2$  is proportional to the concentration of  $N_2O_5$

$$\Rightarrow \text{rate} = \frac{d[O_2]}{dt} = k_2 [NO_3] [NO_2]$$

$$\frac{d[NO_3]}{dt} = 0 = k_f [N_2O_5] - k_b [NO_2] [NO_3] - k_2 [NO_3] [NO_2] + k_3 [NO_3] [NO]$$

$$[NO_3] = \frac{k_f [N_2O_5]}{(k_b + k_2) [NO_2] + k_3 [NO]}$$

$$\text{Rate} = \frac{d[O_2]}{dt} = k_2 [NO_2] \frac{k_f [N_2O_5]}{(k_b + k_2) [NO_2] + k_3 [NO]}$$

at steady state condition  $[NO] = 0$

$$= k_2 k_f \frac{[N_2O_5]}{(k_b + k_2)}$$

∴ Rate formation of  $O_2 \propto [N_2O_5]$

If liquid-vapor equilibrium of an ideal liquid is considered, using clausius equation show that 'the ratio of the molar heat of vaporization of the boiling point is constant'

$$\frac{dp}{dT} = \frac{L}{T(V_{\text{vap}} - V_{\text{liq}})}$$

$$V_{\text{vap}} \gg V_{\text{liq}} \Rightarrow V_{\text{vap}} - V_{\text{liq}} \approx V_{\text{vap}}$$

$$p V_{\text{vap}} = RT$$

$$\Rightarrow V_{\text{vap}} = \frac{RT}{p}$$

$$\frac{dp}{dT} = \frac{L}{T V_{\text{vap}}}$$

$$\frac{dp}{dT} = \frac{L p}{T RT}$$

$$\Rightarrow \frac{dp}{p} = \frac{L}{R} \frac{dT}{T^2}$$

at boiling point  $p = \text{Constant}$  (external pressure)

$$\frac{dp}{dT} = 0 \Rightarrow \frac{L}{T^2} = \text{Constant}$$

$$\Rightarrow \frac{L}{T} = \text{Constant}$$

$$\text{So } \frac{L}{T_b} = \text{Constant}$$

State the physical significance of Clausius Inequality

$$\Rightarrow \text{Clausius inequality } \oint \frac{\delta Q}{T} \leq 0$$

$$1. \text{ Or } \oint \frac{\delta Q}{T} = 0 \rightarrow \text{reversible process}$$

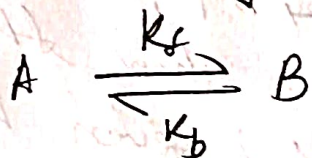
$$< 0 \rightarrow \text{Irreversible (real) process}$$

Heat flows naturally from hot to cold.

Reversible is impossible without external work

It shows entropy of an isolated system never decreases.

for the following elementary reversible reaction:



Derive a rate expression for the overall reaction considering both forward and backward process is a first order reaction.

Forward rxn  $r_f = k_f [A]$

Backward rxn  $r_b = k_b [B]$

$$\text{rate} = \frac{d[B]}{dt} = r_f - r_b = k_f [A] - k_b [B]$$

Let initial concentration  $A = A_0$

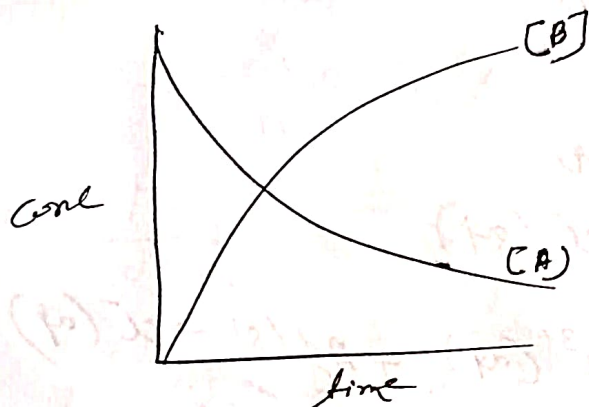
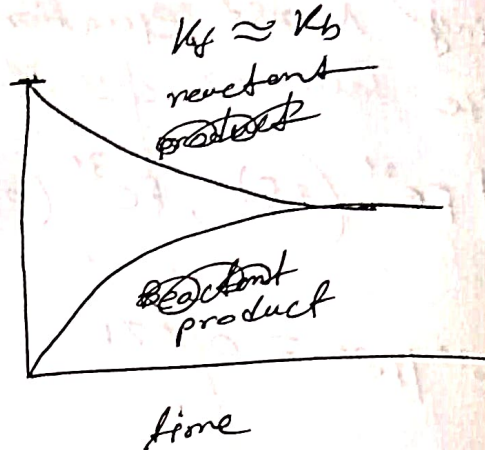
So  $[A] = A_0 - [B]$

$$\begin{aligned} \frac{d[B]}{dt} &= k_f (A_0 - [B]) - k_b [B] \\ &= k_f A_0 - (k_f + k_b) [B] \end{aligned}$$

At eq.

$$K_f[A] = K_b[B]$$

$$\Rightarrow \frac{[B]}{[A]} = \frac{K_f}{K_b} = K_{eq} \quad \text{conc.}$$



The molar Gibbs free energy of a certain gas is given by  $\bar{G} = RT \ln p + a + bT$ , where  $a$  and  $b$  are numerical constant. Show that the equation of state of the gas is given by

$$\bar{V} = RT$$

$$\text{use } d\bar{G} = -s dT + \bar{v} dp$$

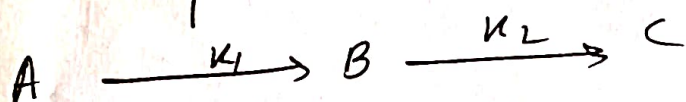
$$\bar{G} = RT \ln p + a + bT$$

$$\begin{aligned} d(\bar{G}) &= d(RT \ln p + a + bT) \\ &= R \ln p dT + RT \frac{dp}{p} + b dT \end{aligned}$$

$$-s dT + \bar{v} dp = (R \ln p + b) dT + \frac{RT}{p} dp$$

$$\Rightarrow \bar{v} = \frac{RT}{p} \quad \text{and} \quad -s = R \ln p + b$$

Show mathematically that for a consecutive reaction the slowest step is the rate determining step



$$-\frac{d[A]}{dt} = k_1[A]$$

$$\frac{d[B]}{dt} = k_1[A] - k_2[B]$$

$$\frac{d[C]}{dt} = k_2[B]$$

Case - 1  $\rightarrow k_1 \ll k_2$  first step slow

B is consumed very fast

$$\frac{d[B]}{dt} \approx 0 \Rightarrow k_1[A] = k_2[B]$$

$$\frac{d[C]}{dt} = k_2[B] = k_1[A]$$

rate depends on  $k_1$

Case - 2  $k_2 \ll k_1$  Second Step Slow

B accumulates (first step fast)

Rate of formation of C

$$\frac{d[C]}{dt} = k_2[B]$$

$\frac{d[C]}{dt} \propto k_2$  overall rate is limited by  $k_2$

Rate of overall reaction is depend on the ~~step~~  
Slow step